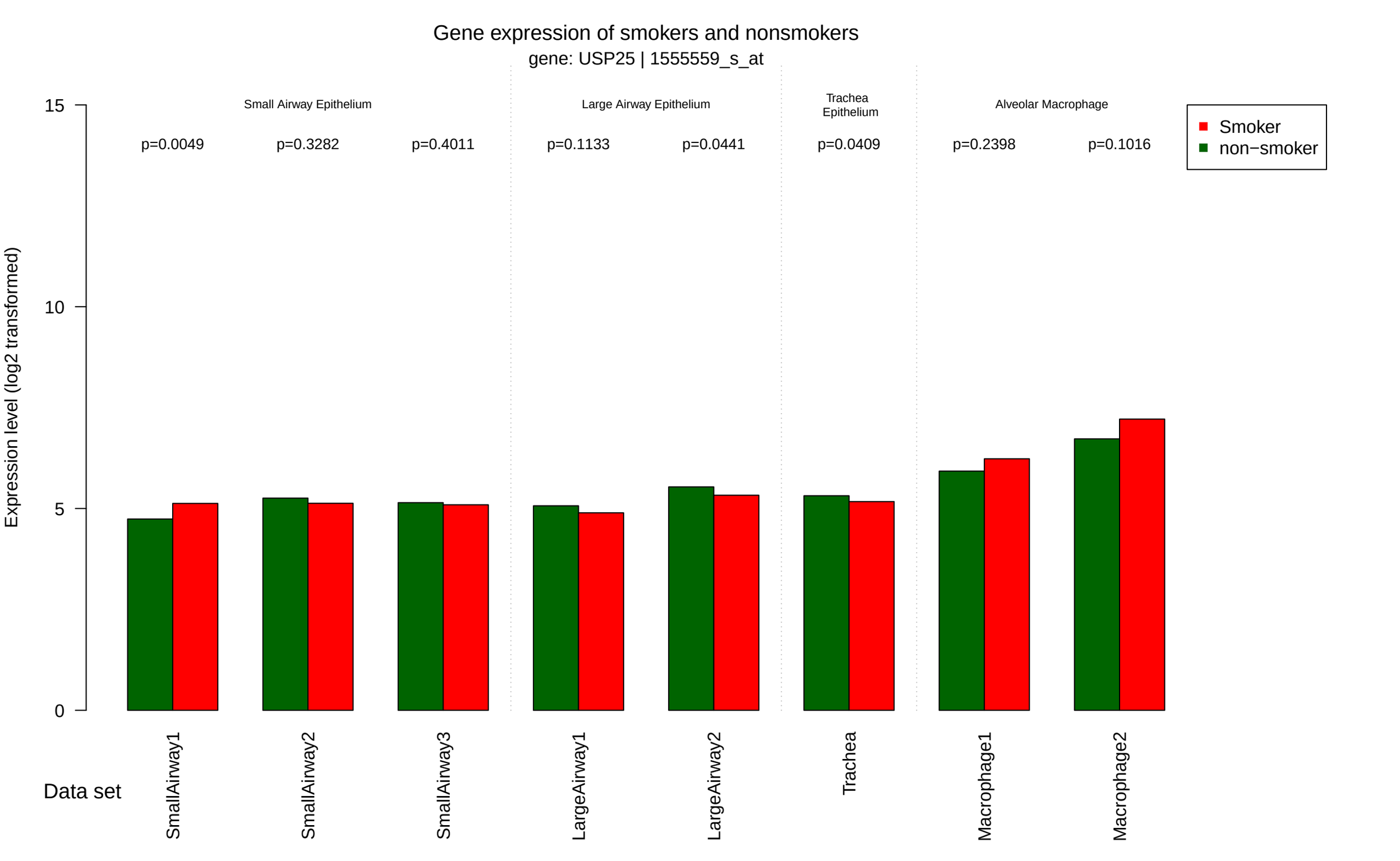
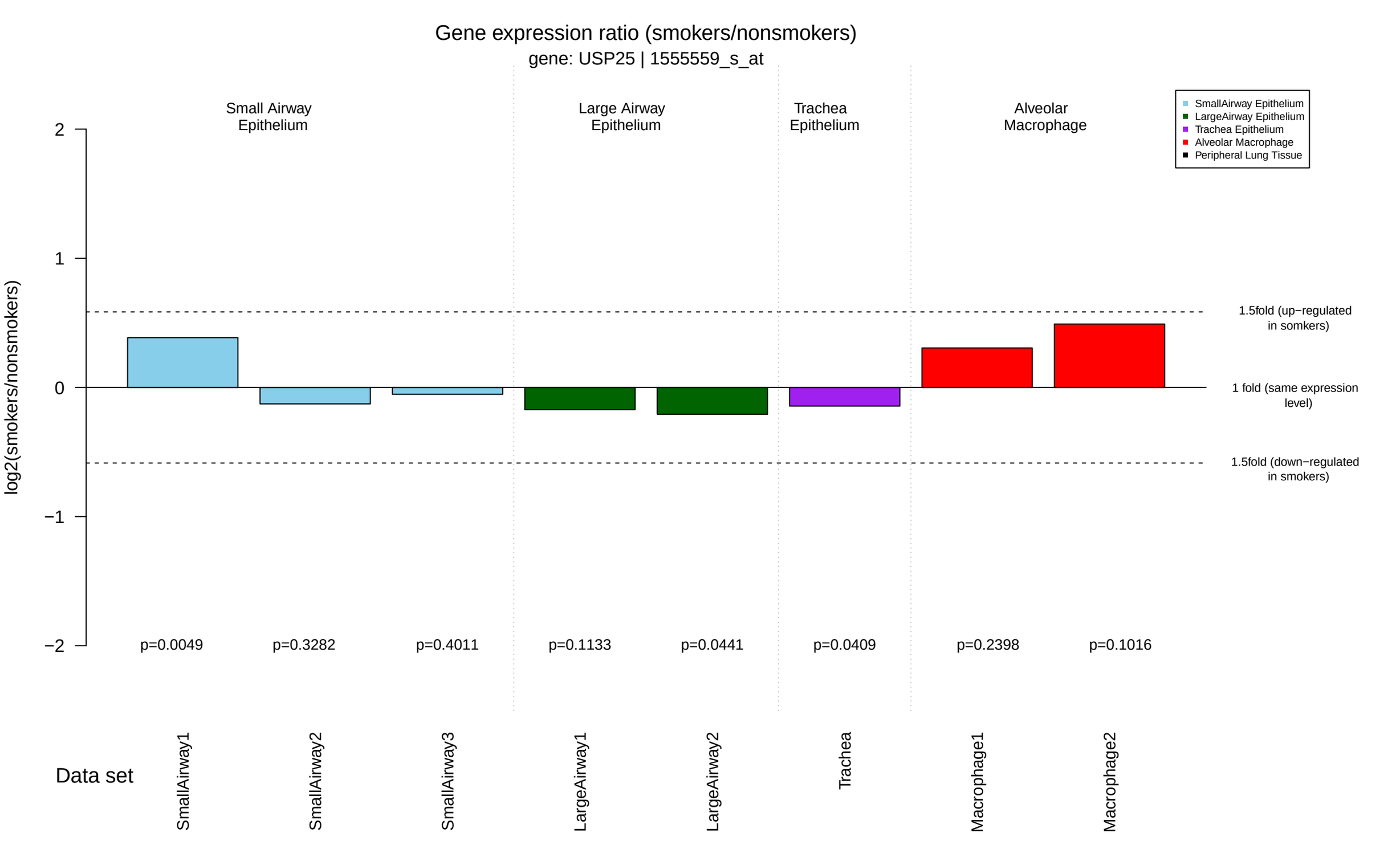
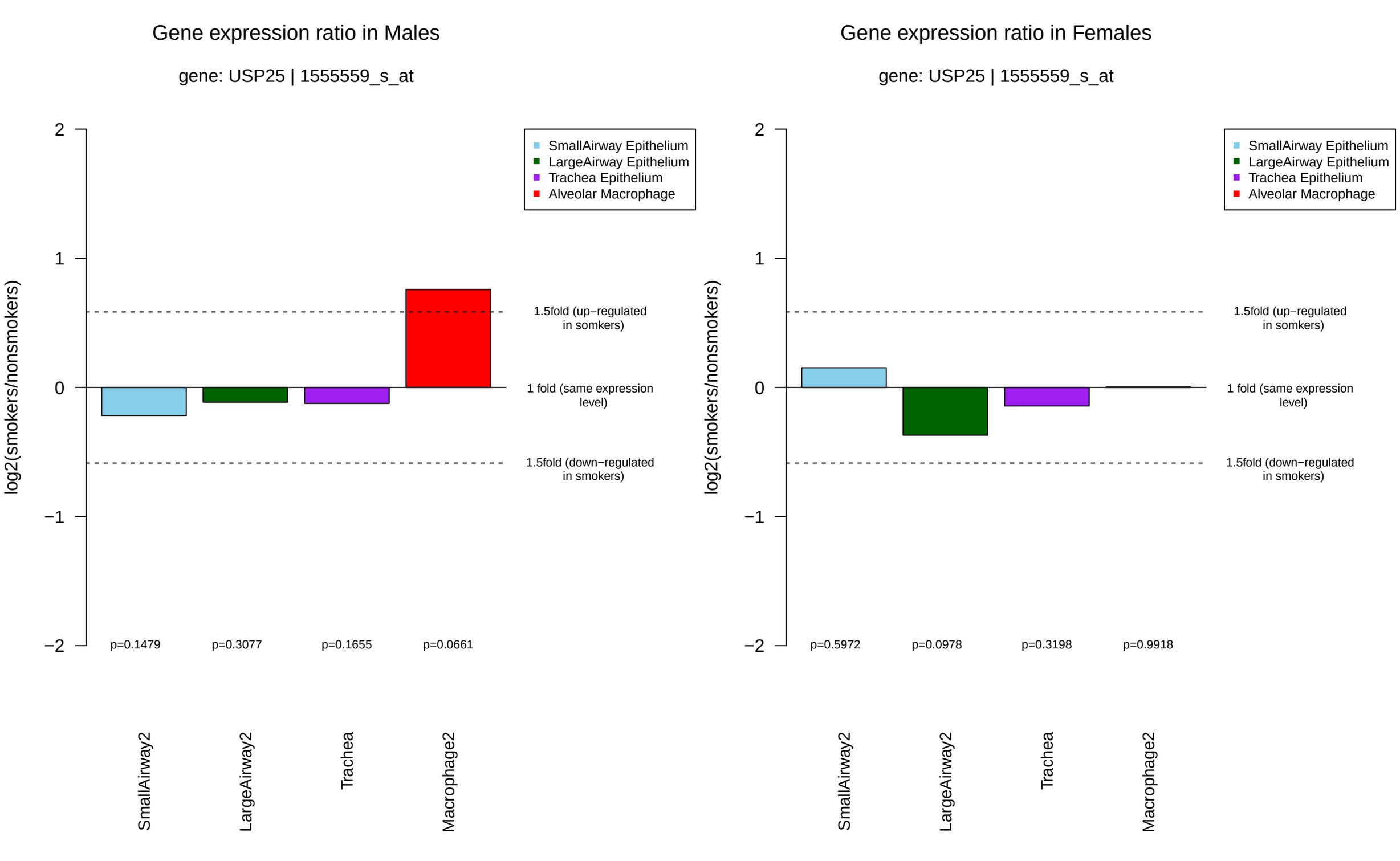
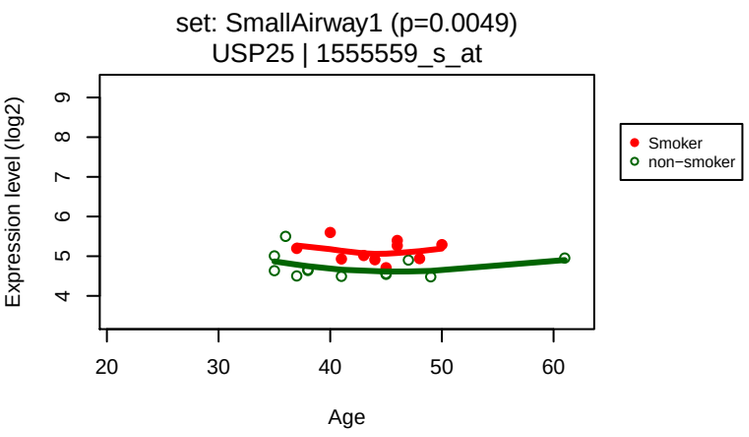
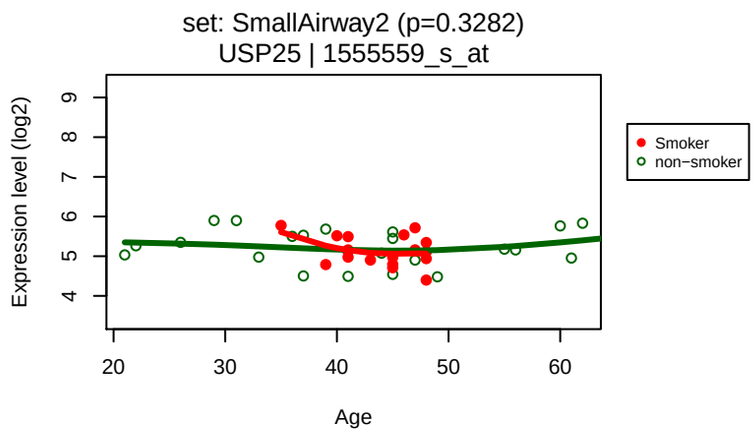
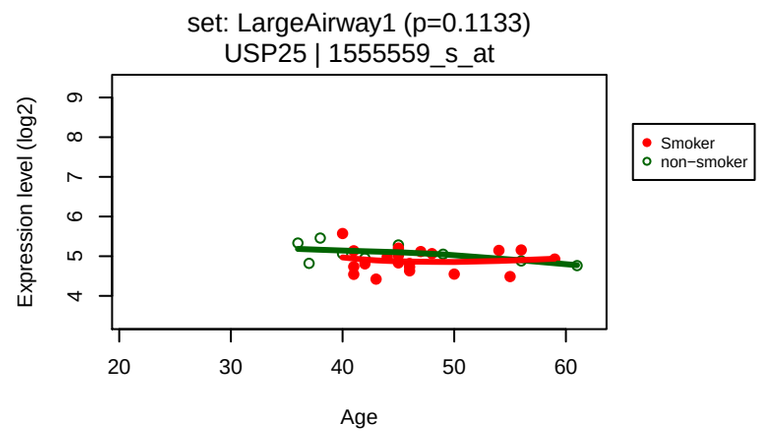

Gene expression of smokers and nonsmokers
gene: USP25 | 1555559_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0049
p=0.3282
p=0.4011
p=0.1133
p=0.0441
p=0.0409
p=0.2398
p=0.1016
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

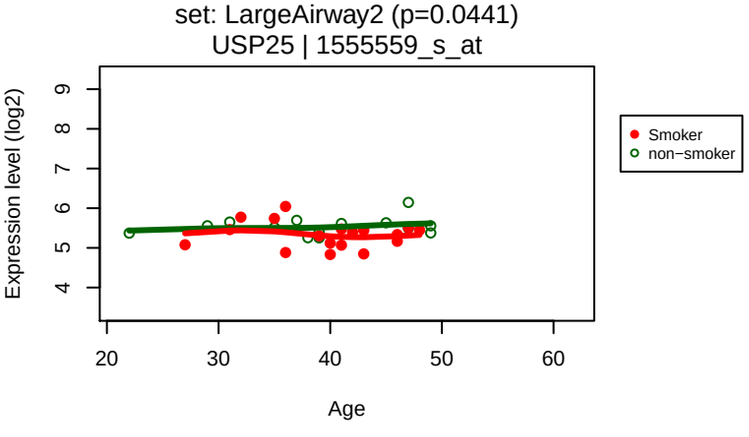

Gene expression ratio (smokers/nonsmokers)
gene: USP25 | 1555559_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0049
p=0.3282
p=0.4011
p=0.1133
p=0.0441
p=0.0409
p=0.2398
p=0.1016
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

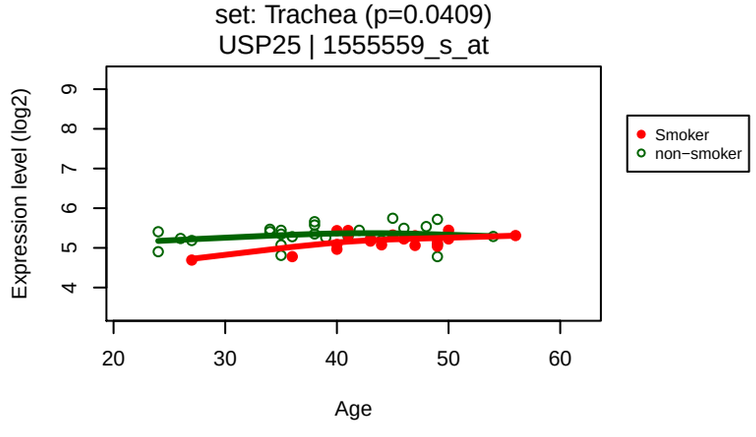

Gene expression ratio in Males
gene: USP25 | 1555559_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1479
p=0.3077
p=0.1655
p=0.0661
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP25 | 1555559_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5972
p=0.0978
p=0.3198
p=0.9918
SmallAirway2
LargeAirway2
Trachea
Macrophage2

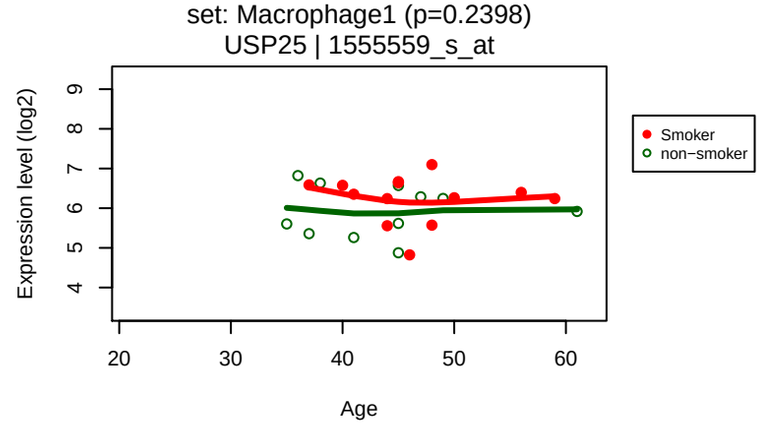

set: SmallAirway1 (p=0.0049)
USP25 | 1555559_s_at
Expression level (log2)
Age
Smoker
non−smoker

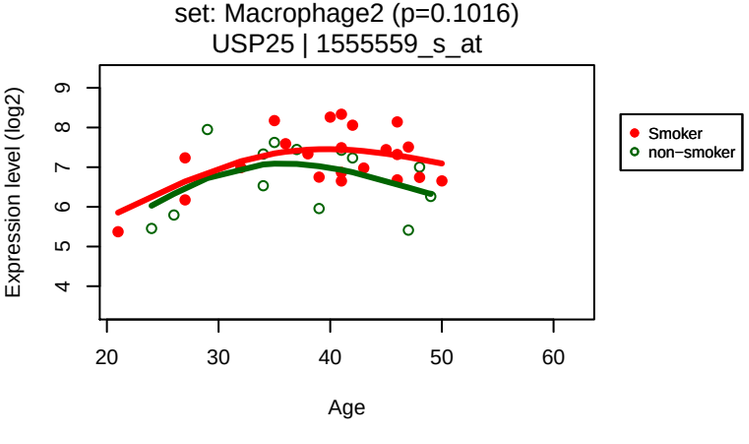

set: SmallAirway2 (p=0.3282)
USP25 | 1555559_s_at
Expression level (log2)
Age
Smoker
non−smoker

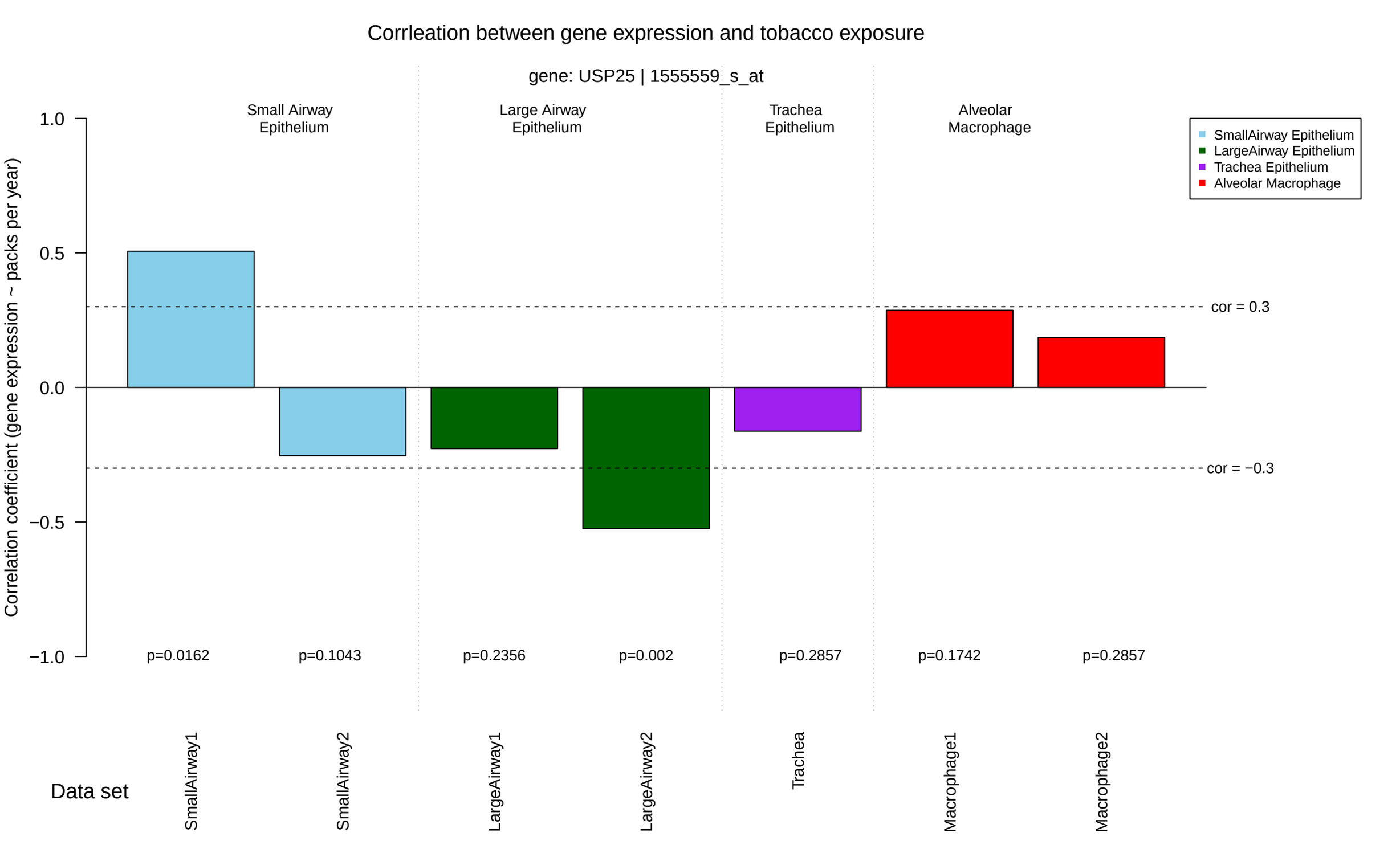

set: LargeAirway1 (p=0.1133)
USP25 | 1555559_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0441)
USP25 | 1555559_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0409)
USP25 | 1555559_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2398)
USP25 | 1555559_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1016)
USP25 | 1555559_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP25 | 1555559_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0162
p=0.1043
p=0.2356
p=0.002
p=0.2857
p=0.1742
p=0.2857
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2